Data Analyst Interview Q&A

Expanded Bilingual Edition — English + Hinglish Explanations

Power BI • DAX • SQL • Data Modeling • ETL / Data Warehousing • Python / Pandas • APIs • PySpark • Microsoft Fabric

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Har jawaab English mein poora diya gaya hai, aur uske saath ek Hinglish (Hindi Roman script mein) box hai jo concept ko seedhi-saadi bhasha mein samjhaata hai.

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1. Power BI Basics

1. What is Power BI used for?

ANSWER

Power BI is Microsoft's business intelligence and data visualization platform. It lets you connect to data from many different sources, clean and reshape that data, build a relational data model, write calculations in DAX, and present the results as interactive reports and dashboards.

EXAMPLE

A retail company connects Excel and SQL Server data to Power BI and builds a report showing total sales, profit margin, trend, top products, and sales by region.

DEEPER DETAIL

Power BI is really three products: Desktop (build), Service (publish/share/refresh), and Mobile (consume on the go). Interviewers want to hear it's an end-to-end pipeline, not just “making charts.”

> *“Power BI is a Microsoft BI tool used to connect, transform, model, and visualize data. I use Power Query for cleaning, DAX for calculations, build interactive reports, and publish them to Power BI Service.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Power BI ek Microsoft ka tool hai jisse aap kisi bhi source (Excel, SQL, cloud) se data connect karke, use clean/shape karke, ek data model banake, DAX se calculations likhke, aur end mein interactive reports/dashboards bana sakte ho. Simple bhasha mein — ye poora pipeline hai: data → clean → model → visual, sirf chart banana nahi hai.

2. What steps do you follow to build a Power BI report?

ANSWER

Connect to the source, clean and shape data in Power Query, build the data model and relationships, write DAX measures, design visuals and layout, validate numbers against the source, and finally publish and schedule refresh.

EXAMPLE

Get Data from SQL Server → clean duplicates/date types in Power Query → create relationships (Sales, Product, Customer, Date) → write measures → build Overview page → cross-check totals → publish.

DEEPER DETAIL

A common mistake is jumping to visuals before the model is solid. Validating numbers before publishing shows attention to data accuracy, not just design.

> *“My usual flow is Connect → Clean → Model → DAX → Visualize → Validate → Publish.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Report banane ka sahi tareeka hai: pehle source se data connect karo, Power Query mein clean karo, phir tables ke beech relationships bana ke ek proper model banao, DAX measures likho, visuals design karo, numbers ko source se match karke validate karo, aur last mein publish + refresh schedule karo. Seedha visuals pe kood mat jao — pehle model solid hona chahiye.

3. Dashboard vs Report

ANSWER

A report is built for detailed, multi-page analysis with many interactive visuals and filters. A dashboard is a single-page, high-level monitoring view made of “tiles” pinned from one or more reports, mainly for at-a-glance KPI tracking.

EXAMPLE

A sales report can have separate pages for Overview, Products, Customers, Regions with drill-through. A dashboard might show only four tiles: Total Sales, Profit, Orders, and a trend chart.

DEEPER DETAIL

Dashboards exist only in Power BI Service (not Desktop), and pinned tiles are static snapshots — clicking one takes you back into the underlying report for real interactivity.

> *“A report is mainly for detailed analysis and can have multiple pages, while a dashboard is mainly for high-level monitoring and quick insights.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Report ek detailed, multi-page cheez hoti hai jisme aap deep-dive kar sakte ho, filters/slicers ke saath. Dashboard ek single-page “quick glance” view hoti hai — chand tiles jo alag-alag reports se pin ki jaati hain, sirf high-level KPIs dekhne ke liye. Yaad rakho: dashboard sirf Power BI Service mein hi bante hain, Desktop mein nahi.

4. What is a dataset / semantic model?

ANSWER

A semantic model (formerly “dataset”) is the data model behind a report — tables, relationships, calculated columns, measures and business logic that visuals pull from. It's the reusable “brain” one or more reports can be built on.

EXAMPLE

A Sales semantic model contains Sales, Product, Customer and Date tables plus measures like Total Sales and YTD Sales. Multiple reports can point to this same model.

DEEPER DETAIL

Because a semantic model can be reused across reports, keeping business logic (measures) centralized rather than duplicated is a best practice — it avoids inconsistent numbers.

> *“A semantic model is the structured model behind a Power BI report. It contains data tables, relationships, measures, and business logic that visuals use for analysis.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Semantic model (pehle isse dataset kehte the) ka matlab hai report ke peeche ka “dimaag” — saari tables, unke relationships, aur measures jo visuals use karte hain. Ek hi model ko multiple reports use kar sakte hain, isliye logic ek jagah rakhna best practice hai taaki alag-alag reports mein numbers mismatch na ho.

5. What is Power Query used for?

ANSWER

Power Query is Power BI's data preparation engine. It connects to sources and lets you clean and reshape data before it reaches the model — removing duplicates, handling nulls, fixing data types, splitting/merging columns, filtering, and merging/appending tables.

EXAMPLE

An Excel export has duplicate rows and a text-formatted date column. In Power Query you remove duplicates on Customer ID and convert the date column's type to Date.

DEEPER DETAIL

Every transformation is recorded as an “Applied Step” and generates M code behind the scenes — knowing Power Query runs on the M language shows depth beyond surface familiarity.

> *“I use Power Query mainly for data extraction, cleaning, and transformation before modeling.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Power Query wo jagah hai jahan aap raw data ko model mein daalne se pehle saaf karte ho — duplicates hatana, nulls handle karna, data types theek karna, columns split/merge karna. Har step “Applied Steps” mein record hota hai aur peeche M language ka code generate hota hai.

6. Measures vs Calculated Columns

ANSWER

A calculated column is computed row by row using row context and is physically stored in the model. A measure is computed on the fly when placed in a visual and responds dynamically to filter context.

EXAMPLE

SalesAmount = Quantity × UnitPrice as a calculated column stores one value per row. Total Sales = SUM(Sales[SalesAmount]) as a measure recalculates depending on filters selected.

DEEPER DETAIL

Calculated columns increase model size/refresh time since they're stored; measures don't. Default to measures for aggregations, use calculated columns only when you need a value to slice/filter/group by.

> *“I use a calculated column when I need a value at row level, while I use a measure for dynamic aggregations and KPIs that should respond to filters.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Calculated column har row ke liye ek fixed value store karta hai (jaise ek naya column jodna). Measure filter ke hisaab se real-time calculate hota hai, storage nahi leta. Aggregation/KPI ke liye measure use karo; row-level value chahiye to hi calculated column.

7. Import vs DirectQuery

ANSWER

In Import mode, Power BI loads a compressed copy of data into its in-memory engine (VertiPaq) — fast but only as fresh as last refresh. In DirectQuery, no data is imported; Power BI sends live queries to the source every interaction.

EXAMPLE

A nightly-refreshed finance report is fine on Import. A live operations dashboard needing orders from the last five minutes uses DirectQuery.

DEEPER DETAIL

DirectQuery trades performance/DAX flexibility for freshness and puts load on the source. Composite models let you mix Import and DirectQuery tables in one model.

> *"Import means the data comes into Power BI's own engine; DirectQuery means Power BI queries the source live every time."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Import mode mein data Power BI ke andar copy ho jaata hai — fast hota hai lekin data sirf last refresh tak fresh rehta hai. DirectQuery mein data import nahi hota, har baar source se live query jaati hai — real-time hota hai lekin thoda slow aur source pe load daalta hai.

8. What is a slicer?

ANSWER

A slicer is an on-canvas, interactive filter visual. Clicking a value updates every other connected visual on the page immediately.

EXAMPLE

A Year slicer with 2024/2025: selecting 2025 filters every chart on the page to show only 2025 data.

DEEPER DETAIL

Slicers can be synced across pages so a selection persists as users navigate — useful for multi-page report UX.

> *"A slicer is a visual filter that allows users to interactively select values and filter the report visuals."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Slicer ek visual filter hai jo report par dikhta hai — user click karke koi value select karta hai (jaise year ya region) aur us page ke saare charts turant us selection ke hisaab se update ho jaate hain.

9. How do you publish a report?

ANSWER

Finish and validate the report in Desktop, then use File → Publish to push it to a workspace in Power BI Service, verify it online, set up scheduled refresh (and a gateway if on-premises), and confirm access.

DEEPER DETAIL

If the source is on-premises, an On-premises Data Gateway is required so Service can reach it for scheduled refresh — shows real deployment experience.

> *"I build and validate the report in Desktop, publish it to the required workspace, and then configure refresh and access as needed."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Publish karne ke liye pehle Desktop mein report ko validate karo, phir File → Publish se use Power BI Service ke workspace mein bhejo. Uske baad refresh schedule karo (agar on-premises source hai to gateway lagega) aur sahi logo ko access do.

10. What is refresh? Why does it fail?

ANSWER

Refresh pulls the latest data from the source into the semantic model. It can fail due to broken connectivity, expired/incorrect credentials, gateway issues, a renamed/deleted source column, or a Power Query step that no longer matches the source shape.

EXAMPLE

A scheduled refresh fails because IT renamed a column from CustName to CustomerName — the Power Query step referencing the old name breaks.

DEEPER DETAIL

Troubleshooting order: read the exact error → verify connectivity/credentials → check gateway status → confirm source object still exists → walk through Power Query steps.

> *"Refresh means updating the model with the latest source data. If it fails, I check connectivity, credentials, gateway, source changes, and transformation errors."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Refresh matlab model ko latest data se update karna. Fail hone ke common reasons: connection tootna, password/credentials expire hona, gateway issue, ya source mein column ka naam change ho jaana jisse Power Query ka step break ho jaata hai. Error message pehle padho, wahi failing step batata hai.

11. Name 3 visuals you used and why

ANSWER

Bar/column charts for comparing values across categories. Line charts for trends over time. Cards/KPI visuals for surfacing one important number at a glance.

EXAMPLE

An Overview page combines a card for Total Revenue, a line chart for 12-month trend, and a bar chart ranking top 10 products.

DEEPER DETAIL

Also worth knowing: matrix visuals for pivot-style cross-tabulation, and maps for geography. Choosing the right visual for comparison vs trend vs part-to-whole is itself a signal.

> *"I commonly use bar charts for category comparisons, line charts for time trends, and cards for important KPIs."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Bar chart categories compare karne ke liye, line chart time ke saath trend dikhane ke liye, aur card ek single important number (jaise Total Sales) dikhane ke liye use hota hai. Sahi visual choose karna hi interview mein achha signal hota hai.

12. What are the different ways to connect to data in Power BI?

ANSWER

Power BI supports files (Excel, CSV, PDF), databases (SQL Server, MySQL, PostgreSQL), cloud services (Azure, SharePoint, Dataverse), and generic connectors like Web, OData, ODBC.

EXAMPLE

A company pulls budget from Excel, transactions from SQL Server, and marketing spend from a Web API — all combined into one model.

DEEPER DETAIL

Some connectors support DirectQuery, some only Import, and some (like folders of CSVs) need a combine/append step first.

> *"Power BI can connect to files, databases, cloud services, and web/API sources, and I choose the connector based on where the data actually lives."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Power BI mein data kai jagah se aa sakta hai — files (Excel, CSV), databases (SQL Server), cloud services (Azure, SharePoint), ya generic Web/API connectors. Data kahan hai uske hisaab se sahi connector choose karo.

13. What is Row-Level Security (RLS) in Power BI?

ANSWER

RLS restricts which rows of data a user can see based on identity, using DAX filter roles defined in the model rather than duplicating the report.

EXAMPLE

A Sales Region role with Region = USERPRINCIPALNAME() ensures a regional manager only sees their region's rows, using the same report everyone uses.

DEEPER DETAIL

RLS is defined once in Desktop (Modeling → Manage Roles) and users are assigned to roles in Service — no need for separate reports per region.

> *"Row-Level Security lets me apply a DAX filter role so different users see only the rows relevant to them, from a single shared report."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
RLS se aap ek hi report bana ke alag-alag users ko sirf unse related data dikha sakte ho — jaise Delhi manager ko sirf Delhi ka data dikhe. Ye DAX filter role se hota hai, alag-alag reports banane ki zaroorat nahi.

14. What is a Power BI Gateway and when do you need one?

ANSWER

An On-premises Data Gateway is software that securely bridges Power BI Service (cloud) and data sources inside a private network, like an on-premises SQL Server.

EXAMPLE

Without a gateway, Power BI Service in the cloud can't reach a sales database sitting inside a company's office network for scheduled refresh.

DEEPER DETAIL

Two modes: Standard gateway (admin-installed, shared) and Personal gateway (single user, Import-only). Fully cloud sources like Azure SQL usually don't need one.

> *"A gateway is required whenever Power BI Service needs scheduled access to a data source that sits behind a company's private network."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Gateway ek pull chain jaisa hai jo Power BI Service (cloud) ko company ke andar (on-premises) rakhe SQL Server tak pahunchne deta hai, taaki scheduled refresh ho sake. Agar source pehle se cloud mein hai (jaise Azure SQL), to gateway ki zaroorat nahi.

15. What are bookmarks and drill-through in Power BI?

ANSWER

A bookmark captures the current state of a page (filters, selections, visibility) so a button can snap back to it, used for guided navigation. Drill-through lets a user right-click a data point to jump to a detail page filtered to that value.

EXAMPLE

A bookmark toggles between 'Summary' and 'Detailed' views. A drill-through page lets a user right-click a product and land on a Product Detail page filtered to it.

DEEPER DETAIL

Bookmarks are often combined with buttons and the Selection pane to build custom navigation menus without separate pages for every state.

> *"Bookmarks save and restore a specific report state for guided navigation, while drill-through jumps to a filtered detail page based on the value a user clicked."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Bookmark ek report page ki current state (filters, selections) save karke rakhta hai, taaki button click karke wahi state wapas aa jaaye. Drill-through se user kisi point pe right-click karke uski detail page pe pahunch jaata hai, already usi value pe filtered.

16. What is a calculation group in Power BI?

ANSWER

A calculation group lets you define reusable calculations like Current, MTD, YTD, or Prior Year, applied to whichever base measure a user picks, instead of writing a separate measure for every base measure.

EXAMPLE

Without calculation groups, 10 base measures needing YTD need 10 separate measures. One calculation group with a YTD calculation item applies to all 10 automatically.

DEEPER DETAIL

Calculation groups are built in Tabular Editor (external tool) rather than Desktop directly — mentioning them signals experience beyond the basic UI.

> *"A calculation group avoids duplicating measures by letting one set of calculations, like YTD or Prior Year, apply dynamically to any base measure a user selects."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Calculation group ek smart shortcut hai — baar baar har measure ke liye alag YTD/MTD banane ke bajaye, ek hi calculation group bana ke usse kisi bhi measure pe apply kar sakte ho. Ye Tabular Editor (external tool) mein banta hai.

17. How do you optimize a slow Power BI report?

ANSWER

Look at it in layers: model size/structure first (unused columns, Import over DirectQuery, star schema, integer keys), then DAX efficiency (avoid unnecessary iterators, use variables), then visual-level design.

DEEPER DETAIL

Use Performance Analyzer to actually measure bottlenecks instead of guessing — it shows exactly how long each visual and DAX query took.

> *"I use Performance Analyzer to find the actual bottleneck, then optimize in order: the data model, the DAX, and finally the visual layout, rather than guessing."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Report slow hone par pehle model dekho (extra columns hatao, star schema use karo), phir DAX optimize karo (variables use karo, unnecessary iterators mat lagao), aur last mein visuals ka layout dekho. Guess mat karo — Performance Analyzer tool se exact bottleneck pata karo.

2. DAX Interview Questions

18. SUM vs CALCULATE

ANSWER

SUM is a simple aggregation that adds values in one column. CALCULATE evaluates an expression inside a modified filter context — it lets you override or add filters before an aggregation runs.

EXAMPLE

Total Sales = SUM(Sales[SalesAmount]) sums visible rows. US Sales = CALCULATE([Total Sales], Sales[Country]="USA") forces the calculation to only consider USA rows.

DEEPER DETAIL

CALCULATE is the single most important DAX function — nearly every advanced pattern (YTD, previous year, running totals) is built by wrapping an aggregation inside it.

> *"SUM performs a straightforward aggregation, while CALCULATE is used when I need to modify filter context for a calculation."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

SUM sirf ek column ke values jod deta hai. CALCULATE uss calculation ke filters ko badal/add kar sakta hai — jaise "USA ka sales nikaalo, chahe visual mein koi bhi filter lage ho." CALCULATE DAX ka sabse important function hai.

19. Row Context vs Filter Context

ANSWER

Row context is DAX evaluating one row at a time (calculated columns, iterators like SUMX). Filter context is the set of filters currently applied, coming from slicers, visual fields, page/report filters, or CALCULATE.

EXAMPLE

Profit = Sales[SalesAmount] - Sales[Cost] as a calculated column uses row context. A Total Sales measure in a table by Region uses filter context.

DEEPER DETAIL

Row context does NOT automatically apply filters like filter context does — this is why 'context transition' exists, to convert a row into an equivalent filter.

> *"Row context evaluates data row by row, while filter context determines which data is included in a calculation."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Row context matlab ek row pe ek time pe calculation (jaise calculated column mein). Filter context matlab ki kaunse filters currently laage hain (slicer, visual, page filter). Dono alag hain — row context automatically filter nahi banata, isliye CALCULATE ki zaroorat padti hai.

20. SUM vs SUMX

ANSWER

SUM aggregates an existing numeric column directly. SUMX is an iterator: for each row it evaluates an expression, then adds up all per-row results.

EXAMPLE

SUM(Sales[SalesAmount]) vs SUMX(Sales, Sales[Quantity] * Sales[UnitPrice]).

DEEPER DETAIL

Use SUM when the number already exists as a column. Use SUMX when the value must be computed per row first before summing — a plain SUM can't do that multiplication.

> *"I use SUM when the value already exists in a column, and SUMX when I need a row-level calculation performed before the results are aggregated."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

SUM sirf ek existing column ko jodta hai. SUMX pehle har row ke liye ek calculation karta hai (jaise Quantity × Price), phir sabko jodta hai. Jab value pehle se column mein nahi hai aur banani padti hai, tab SUMX use hota hai.

21. What is context transition?

ANSWER

Context transition is what happens when CALCULATE converts the current row context into an equivalent filter context. Measures normally only react to filter context, so without it a row context alone couldn't drive a measure's result.

EXAMPLE

Inside SUMX(Customers, CALCULATE([Total Sales])), for each customer row, CALCULATE turns that row into a filter on Customer, so [Total Sales] returns only that customer's sales.

DEEPER DETAIL

This is a favorite interview trick because it's the root cause of a common bug: using a measure directly inside an iterator without CALCULATE gives the overall total on every row.

> *"Context transition is the process where CALCULATE turns the current row context into filter context, which is what lets a measure calculate correctly inside an iterator."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Context transition ek trick hai jahan CALCULATE current row ko ek filter mein badal deta hai. Isse ek measure iterator ke andar sahi se kaam karta hai — har customer ke liye sirf uska data aata hai, poora total nahi.

22. What is YTD Sales?

ANSWER

YTD (Year To Date) is a cumulative total from the first day of the current year up to whatever date is currently in context. YTD Sales = TOTALYTD([Total Sales], 'Date'[Date])

EXAMPLE

If filtered to March 31, YTD Sales adds every day's sales from January 1 through March 31 of that year.

DEEPER DETAIL

TOTALYTD requires a proper marked Date table with a continuous, unbroken range of dates — gaps will silently break YTD/MTD/QTD calculations.

> *"YTD calculates the cumulative value from the start of the year through the current date."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

YTD ka matlab hai saal ki shuruwat (1 January) se leke abhi tak ka total. Isko sahi kaam karne ke liye ek proper Date table honi zaroori hai jisme koi gap na ho.

23. What is MoM Growth %?

ANSWER

Month-over-Month growth compares the current month's value against the immediately preceding month, as a percentage change.

EXAMPLE

If January sales are 100 and February sales are 120, MoM growth is (120-100)/100 = 20%.

DEEPER DETAIL

DIVIDE is preferred over the / operator since it gracefully returns blank instead of a divide-by-zero error.

> *"MoM growth measures how much the current month increased or decreased compared with the previous month."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

MoM Growth batata hai is mahine ka sales pichle mahine se kitna percent zyada ya kam hua. Formula mein DIVIDE use karo, / nahi — warna zero se divide hone par error aa sakta hai.

24. What is a Running Total?

ANSWER

A running total (cumulative total) is the sum of all values up to and including the current point in an ordered sequence.

EXAMPLE

Daily sales of 10, 20 and 30 produce running totals of 10, 30 and 60.

DEEPER DETAIL

Useful for cumulative sales/orders/profit charts where the shape of accumulation over time matters more than any single day's value.

> *"A running total accumulates values up to the current point in an ordered sequence, like cumulative sales over time."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Running total matlab abhi tak ka poora jod — har din ka number pichle sab din ke sath mila ke dikhaya jaata hai. Isse cumulative trend dikhta hai.

25. What is Rolling 7 Days?

ANSWER

A rolling (moving) 7-day calculation looks at a sliding window of the current day plus the previous six days, recalculating for every new day.

EXAMPLE

The value shown for July 10 covers July 4–10; for July 11 it shifts to July 5–11.

DEEPER DETAIL

Useful for smoothing out noisy daily fluctuations (weekday vs weekend dips) so the underlying trend is easier to read.

> *"A rolling 7-day measure calculates a moving total or average over the current day and the previous six days, which helps smooth daily fluctuations."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Rolling 7 days ka matlab hai har din ke liye pichle 7 dino ka average/total nikaalna — jaise ek chalta hua window. Isse daily ups-downs smooth ho jaate hain aur trend saaf dikhta hai.

26. ALL / ALLEXCEPT / ALLSELECTED

ANSWER

ALL removes all filters from a table/columns. ALLEXCEPT removes all filters except the ones you keep. ALLSELECTED removes filters from the current visual's own context but respects filters applied outside it (slicers, page filters).

EXAMPLE

% of Total Sales = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Sales))) always divides by the grand total.

DEEPER DETAIL

ALLSELECTED is the trickiest — used when a subtotal should reflect what the user selected via slicer but be independent of the row/column breakdown inside the visual itself.

> *"I use ALL when I need to ignore filters, ALLEXCEPT when I want to preserve selected grouping columns, and ALLSELECTED when I need a calculation that respects the user's selections."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
ALL sab filters hata deta hai. ALLEXCEPT kuch filters ko chhod ke baaki hata deta hai. ALLSELECTED visual ke andar ke filters hata deta hai but bahar ke slicers ko respect karta hai. Teeno 'percent of total' jaisi calculations mein kaam aate hain.

27. How do you debug wrong numbers in DAX?

ANSWER

First confirm what the correct number should be by checking the source or running an equivalent SQL query, then work through relationships, filters/slicers, column data types, and finally the DAX formula, building small isolated test measures.

DEEPER DETAIL

Checklist: Source → Relationships → Filters → Data Types → DAX → Visual. Bidirectional/many-to-many relationships are common culprits for double-counting.

> *"I compare the Power BI result with the source, then isolate the issue by checking relationships, filters, data types, and smaller test measures."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Number galat lage to seedha DAX formula change mat karo. Pehle source data se compare karo ki sahi number kya hona chahiye. Fir step by step check karo: relationships, filters, data types, aur last mein DAX formula.

28. What is a variable (VAR) in DAX and why use it?

ANSWER

VAR lets you store the result of an expression under a name and reuse it later via RETURN, instead of repeating the same calculation multiple times.

EXAMPLE

VAR TotalRevenue = SUM(Sales[SalesAmount]); VAR TotalCost = SUM(Sales[Cost]); RETURN DIVIDE(TotalRevenue - TotalCost, TotalRevenue)

DEEPER DETAIL

Variables improve readability and performance: DAX evaluates each variable once and reuses the value, whereas repeating a sub-expression inline can force recomputation.

> *"I use VAR to store intermediate results once, which makes complex DAX measures easier to read, easier to debug, and often faster."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
VAR se aap ek calculation ka result naam de ke store kar sakte ho, aur usse baar baar use kar sakte ho RETURN mein. Isse formula padhna aasan ho jaata hai aur performance bhi better hoti hai.

29. What are Time Intelligence functions in DAX?

ANSWER

Built-in DAX functions that make date-based calculations like YTD, QTD, MTD, previous period, and same-period-last-year easy, given a proper marked Date table.

EXAMPLE

Sales Last Year = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date])); YoY Growth % = DIVIDE([Total Sales]-[Sales Last Year], [Sales Last Year])

HINDI MEIN SAMJHAAIYE (HINGLISH)

Time Intelligence functions (TOTALYTD, SAMEPERIODLASTYEAR, DATEADD, etc.) date-based calculations aasan bana dete hain — lekin inke sahi kaam karne ke liye ek proper, continuous Date table zaroori hai.

3. SQL Interview

30. Write a query using JOIN

ANSWER

JOIN combines related rows from two or more tables based on a matching key column.

EXAMPLE

```
SELECT e.name, d.department FROM employees e INNER JOIN departments d ON e.dept_id = d.dept_id;
```

DEEPER DETAIL

INNER JOIN only returns rows where a match exists in both tables; unmatched rows are dropped entirely.

> *"I use JOIN when related information is stored in different tables and I need to combine it using a common key."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
JOIN do tables ko ek common column (key) ke through jodta hai. INNER JOIN sirf wahi rows deta hai jinka dono tables mein match ho — jinka match nahi hota woh chhoot jaati hain.

31. GROUP BY + HAVING

ANSWER

GROUP BY collapses rows into groups so aggregates like COUNT/SUM/AVG can be calculated per group. HAVING filters those grouped results after aggregation.

EXAMPLE

```
SELECT department, COUNT(*) FROM employees GROUP BY department HAVING COUNT(*) > 2;
```

DEEPER DETAIL

WHERE filters individual rows before grouping; HAVING filters after. You can't use an aggregate function inside WHERE.

> *"GROUP BY creates the groups for aggregation, and HAVING filters those groups after the aggregate values are calculated."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
GROUP BY rows ko groups mein baant deta hai (jaise department ke hisaab se), aur uske baad HAVING un groups ko filter karta hai (jaise sirf wahi department jinme 2 se zyada log hain).

32. How do you find duplicate records?

ANSWER

Group rows by whichever column should be unique, then use HAVING COUNT(*) > 1 to surface any group appearing more than once.

EXAMPLE

```
SELECT name, COUNT(*) FROM employees GROUP BY name HAVING COUNT(*) > 1;
```

DEEPER DETAIL

In a real project, use the actual business key (email, ID) rather than name alone, so two different people with the same name aren't flagged.

> *"I group by the column that should be unique and use HAVING COUNT(*) > 1 to find duplicates, using the correct business key for the table."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Duplicates dhundne ke liye us column pe GROUP BY karo jo unique hona chahiye, phir HAVING COUNT(*) > 1 lagao. Real project mein sirf naam nahi, balki email ya ID jaisa sahi unique key use karo.

33. What is ROW_NUMBER?

ANSWER

ROW_NUMBER is a window function that assigns a unique, strictly sequential integer to each row based on the ORDER BY inside OVER(), optionally restarting per group with PARTITION BY.

EXAMPLE

```
SELECT name, salary, ROW_NUMBER() OVER(ORDER BY salary DESC) AS rn FROM employees;
```

DEEPER DETAIL

Even if two employees tie on salary, ROW_NUMBER still gives different numbers unless a tie-breaker is added — the key difference from RANK/DENSE_RANK.

> *"ROW_NUMBER assigns a unique sequential number to each row based on an ORDER BY, and never repeats a number even when values tie."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
ROW_NUMBER har row ko ek unique number deta hai order ke hisaab se (1, 2, 3...). Agar do rows ka value same bhi ho, tab bhi number repeat nahi hota.

34. Primary Key vs Foreign Key

ANSWER

A primary key uniquely identifies each record in its own table and cannot be NULL. A foreign key is a column referencing the primary key of another table, establishing the relationship.

EXAMPLE

department_id is the primary key in Departments and a foreign key in Employees, linking each employee to their department.

DEEPER DETAIL

Foreign keys enforce referential integrity — the database rejects an insert referencing a department_id that doesn't exist.

> *"Primary key identifies a row uniquely; foreign key connects one table to another and enforces that relationship."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Primary key ek table mein har row ko unique identify karta hai (NULL nahi ho sakta). Foreign key doosri table ke primary key ko point karta hai, jisse dono tables aapas mein connect hote hain.

35. WHERE vs HAVING

ANSWER

WHERE filters individual rows before any grouping/aggregation. HAVING filters groups after GROUP BY has aggregated the rows.

EXAMPLE

SELECT department, COUNT(*) FROM employees WHERE salary > 50000 GROUP BY department HAVING COUNT(*) > 2;

DEEPER DETAIL

WHERE removes low-salary employees first; only then does GROUP BY form groups; HAVING then keeps only departments with more than two remaining employees.

> *"WHERE filters rows before grouping; HAVING filters groups after aggregation."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

WHERE grouping se pehle individual rows filter karta hai. HAVING grouping ke baad groups ko filter karta hai. Dono ka order yaad rakho: pehle WHERE, phir GROUP BY, phir HAVING.

36. INNER JOIN vs LEFT JOIN

ANSWER

INNER JOIN returns only matching rows in both tables. LEFT JOIN returns every row from the left table regardless of a match, filling right-side columns with NULL when there's no match.

EXAMPLE

Customers LEFT JOIN Orders keeps every customer, even those who've never ordered — their order columns show NULL.

DEEPER DETAIL

LEFT JOIN is right whenever the question is about the left-hand entity regardless of activity, e.g. every customer including those with zero orders.

> *"I use INNER JOIN when I only need matching records and LEFT JOIN when all records from the left table must be retained."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

INNER JOIN sirf wahi rows deta hai jo dono tables mein match karti hain. LEFT JOIN left table ki saari rows rakhta hai, chahe right table mein match ho ya na ho (NULL aa jaata hai).

37. Find top 5 products by revenue

ANSWER

Group sales by product, sum revenue per group, sort highest to lowest, keep only the top five.

EXAMPLE

SELECT product_id, SUM(revenue) AS total_revenue FROM sales GROUP BY product_id ORDER BY total_revenue DESC LIMIT 5;

DEEPER DETAIL

LIMIT syntax differs by database — SQL Server uses TOP 5, Oracle uses ROWNUM or FETCH FIRST 5 ROWS ONLY.

> *"I group sales by product, aggregate total revenue, sort descending, and return the first five rows."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Top 5 products nikaalne ke liye: product ke hisaab se group karo, revenue sum karo, descending order mein sort karo, aur sirf top 5 rows lo (LIMIT 5).

38. Count customers per city

ANSWER

Group customers by city and count the rows in each group.

EXAMPLE

```
SELECT city, COUNT(*) AS customer_count FROM customers GROUP BY city;
```

DEEPER DETAIL

If some rows have NULL city, most databases group them as an 'unknown city' bucket — worth flagging to a stakeholder.

> *"GROUP BY creates one group per city, and COUNT(*) gives the number of customers in each group."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Har city ke customers count karne ke liye GROUP BY city lagao aur COUNT(*) use karo. NULL city wali rows apne aap ek alag group ban jaati hain.

39. What is a subquery?

ANSWER

A subquery is a query nested inside another query, used to feed a value, list, or derived table to the outer query.

EXAMPLE

```
SELECT name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
```

DEEPER DETAIL

Subqueries can be correlated or non-correlated; a correlated subquery re-runs per outer row and can be slower than an equivalent JOIN on large tables.

> *"A subquery is a query nested inside another query, often used to supply a calculated value like an average for the outer query to compare against."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Subquery ek query ke andar dusri query hoti hai — jaise pehle average salary nikaalo, phir usse compare karke employees dhundo jo usse zyada kamate hain.

40. What is indexing?

ANSWER

An index is a separate database structure built on one or more columns that lets the engine find matching rows quickly instead of scanning the whole table.

EXAMPLE

```
CREATE INDEX idx_employee_dept ON employees(dept_id);
```

DEEPER DETAIL

Indexes speed up SELECT/JOIN/ORDER BY on indexed columns but add storage overhead and slow down writes since the index must be maintained on every INSERT/UPDATE/DELETE.

> *"Indexing in SQL is a technique that speeds up searching and retrieving data, similar to the index at the back of a book."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Index ek book ke index jaisa hota hai — usse data dhundna fast ho jaata hai, poori table scan nahi karni padti. Lekin har write (insert/update/delete) pe index ko bhi update karna padta hai, isliye zaroori columns pe hi lagao.

41. Normalization vs Denormalization

ANSWER

Normalization organizes data across related tables to reduce duplicate data and data-consistency problems. Denormalization intentionally combines/duplicates data across tables to simplify and speed up read-heavy queries.

EXAMPLE

A normalized system keeps Customers, Orders, Products separate. A denormalized reporting table might repeat the customer's city on every order row.

DEEPER DETAIL

OLTP systems favor normalization (fast, safe writes); OLAP systems favor denormalization (fewer joins, faster reads) — like a Power BI star schema.

> *"Normalization reduces redundancy and keeps data consistent; denormalization can make analytical queries simpler and faster by reducing joins."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Normalization data ko alag-alag related tables mein todta hai taaki repetition na ho aur data consistent rahe. Denormalization jaan-boojh kar data repeat karta hai taaki reports/reads fast ho jaayein — isliye Power BI ke star schema mein denormalization pasand kiya jaata hai.

42. How do you handle NULLs?

ANSWER

First figure out why a value is NULL and what the business rule says. Then filter with IS NULL/IS NOT NULL, replace with a default using COALESCE, or deliberately leave it NULL when absence is meaningful.

EXAMPLE

SELECT COALESCE(phone, 'Not Available') AS phone FROM customers;

DEEPER DETAIL

Avoid blindly replacing every NULL with 0 — NULL (unknown/missing) and 0 (confirmed zero) often carry different business meaning.

> *"I handle NULLs based on business meaning — filtering, defaulting with COALESCE, or leaving them as NULL — rather than assuming one rule fits every column."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
NULL ka matlab samjho pehle — kyun missing hai. Fir decide karo: filter karo, COALESCE se default value do, ya NULL hi rehne do agar wo meaningful hai. NULL aur 0 same nahi hote — jaise missing discount aur 0% discount alag cheez hai.

43. RANK vs DENSE_RANK

ANSWER

Both are window functions assigning rank based on ORDER BY, but treat ties differently. RANK leaves a gap after a tie; DENSE_RANK does not.

EXAMPLE

For salaries 70000, 70000, 60000: RANK returns 1,1,3. DENSE_RANK returns 1,1,2.

DEEPER DETAIL

Use RANK for competition-style ranking where gaps after ties are expected. Use DENSE_RANK when you need consecutive numbers regardless of ties.

> *"RANK skips numbers after a tie; DENSE_RANK keeps ranks consecutive even when values tie."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
RANK aur DENSE_RANK dono rank dete hain, lekin tie hone par RANK number skip kar deta hai (1,1,3), jabki DENSE_RANK skip nahi karta (1,1,2).

44. UNION vs UNION ALL

ANSWER

UNION combines two compatible result sets and removes duplicate rows. UNION ALL combines them but keeps every row, including duplicates.

DEEPER DETAIL

Because UNION compares every row to eliminate duplicates, it's more expensive than UNION ALL. When result sets can't overlap or duplicates are fine, UNION ALL is faster.

> *"UNION removes duplicates; UNION ALL keeps them and is generally faster since it skips the duplicate-elimination step."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
UNION do results ko jodta hai aur duplicates hata deta hai. UNION ALL sab kuch rakhta hai including duplicates, aur isliye faster hota hai.

45. What is a Self Join?

ANSWER

A self join is a join where a table is joined to itself, treated as two separate tables using aliases, used when rows need to be compared to other rows in the same table.

EXAMPLE

SELECT e.name AS employee, m.name AS manager FROM employees e LEFT JOIN employees m ON e.manager_id = m.employee_id;

DEEPER DETAIL

It's called 'self' because both sides reference the same table; aliases let SQL tell the two 'copies' apart. Also used for finding employees earning more than their manager.

> *"A self join is a table joined to itself using aliases, typically used for hierarchical or comparative relationships like employee-to-manager."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Self join mein ek table ko khud ke saath join karte hain (alias use karke) — jaise employee aur uske manager ko dhundna, jab dono ek hi table mein hon.

46. What are Window Functions in SQL?

ANSWER

Window functions perform a calculation across a related set of rows without collapsing them into a single output row like GROUP BY — every original row stays visible with the calculated value alongside it.

EXAMPLE

SELECT name, department, salary, AVG(salary) OVER(PARTITION BY department) AS dept_avg_salary FROM employees;

DEEPER DETAIL

PARTITION BY resets the calculation per group without collapsing rows; ORDER BY inside OVER() controls running/ranking calculations. Common ones: ROW_NUMBER, RANK, SUM/AVG OVER, LAG, LEAD.

> *“Window functions let me calculate things like rankings, running totals, or group averages while still returning every individual row, unlike GROUP BY which collapses rows.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Window functions GROUP BY jaisi calculations karte hain (rank, average) lekin har original row ko rakhte hain, GROUP BY ki tarah rows ko collapse nahi karte.

47. What are LAG and LEAD in SQL?

ANSWER

LAG returns a value from a previous row and LEAD returns a value from a following row, relative to the current row, based on ORDER BY.

EXAMPLE

SELECT month, sales, LAG(sales,1) OVER(ORDER BY month) AS prev_month_sales, sales - LAG(sales,1) OVER(ORDER BY month) AS change FROM monthly_sales;

DEEPER DETAIL

Before window functions, this comparison required a self join on a date offset, which was clunkier — LAG/LEAD is the modern, cleaner way.

> *“LAG looks back to a previous row and LEAD looks ahead to a following row, which makes period-over-period comparisons straightforward without a self join.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
LAG pichhli row ki value laata hai, LEAD agli row ki value laata hai — isse month-over-month jaisi comparisons aasani se ho jaati hain, self join ki zaroorat nahi.

48. What is a CTE (Common Table Expression)?

ANSWER

A CTE is a named, temporary result set defined with WITH that you reference later in the same query, breaking complex SQL into logical, readable steps.

EXAMPLE

WITH dept_totals AS (SELECT department, SUM(salary) AS total_salary FROM employees GROUP BY department) SELECT * FROM dept_totals WHERE total_salary > 500000;

DEEPER DETAIL

A CTE only exists for the duration of that query (unlike a view). A Recursive CTE can reference itself — the standard way to query hierarchical data.

> *“A CTE lets me name and reuse an intermediate result within a single query, which makes multi-step logic far more readable than nesting subqueries.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
CTE (WITH clause) ek temporary named result hota hai jo sirf usi query ke andar use hota hai — isse complex queries padhna aasan ho jaata hai, nested subqueries se better.

49. What is a View in SQL?

ANSWER

A view is a saved, named SELECT query that behaves like a virtual table — it doesn't store data itself; it re-runs its underlying query each time it's referenced.

EXAMPLE

CREATE VIEW active_customers AS SELECT customer_id, name, city FROM customers WHERE status = 'Active';

DEEPER DETAIL

Views hide complexity and help with security (exposing only certain columns/rows). A materialized view stores its result physically and needs refreshing.

> *"A view is a saved query that acts like a virtual table, useful for simplifying complex joins and controlling what data different users can see."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

View ek saved query hai jo virtual table jaisi dikhti hai, lekin data khud store nahi karta — har baar underlying query dobara chalti hai. Complex joins ko simplify karne aur security ke liye useful hota hai.

50. What is a Stored Procedure, and how is it different from a Function?

ANSWER

A stored procedure is a saved, precompiled block of SQL that can be called by name, take parameters, and return multiple result sets. A function must return a single value (or table) and can be used directly inside a SELECT.

EXAMPLE

CREATE PROCEDURE GetEmployeesByDept (IN dept_id INT) BEGIN SELECT * FROM employees WHERE department_id = dept_id; END;

DEEPER DETAIL

Procedures are used for reusable business logic/batch operations; functions are used when you need a computed value plugged directly into a query.

> *"A stored procedure performs an action and can return multiple result sets, while a function returns a single value or table and can be used directly inside a SELECT statement."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Stored Procedure ek saved block of SQL hai jo action perform karta hai aur multiple results de sakta hai. Function ek single value/table return karta hai aur seedha SELECT ke andar use ho sakta hai.

51. What is a Trigger in SQL?

ANSWER

A trigger is a block of SQL that runs automatically in response to a specific event — typically INSERT, UPDATE, or DELETE — on a table, without being called explicitly.

EXAMPLE

An AFTER INSERT trigger on Orders automatically writes a row into OrderAudit every time a new order is inserted.

DEEPER DETAIL

Triggers are powerful for auditing/enforcing rules automatically, but overusing them makes behavior hard to trace since logic fires silently in the background.

> *"A trigger automatically runs SQL logic in response to an INSERT, UPDATE, or DELETE event, commonly used for auditing or enforcing business rules at the database level."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Trigger ek code hai jo apne aap chal jaata hai jab kisi table mein INSERT/UPDATE/DELETE hota hai — jaise order insert hote hi audit table mein entry ban jaana, bina kisi ko explicitly call kiye.

52. What is Data Warehousing?

ANSWER

A data warehouse is a central repository that consolidates data from multiple source systems into one structured, historical store optimized for reporting and analysis rather than day-to-day transactions.

EXAMPLE

Instead of querying the live production database directly, a company loads data nightly into a data warehouse where analysts and Power BI query freely without affecting operations.

DEEPER DETAIL

Data warehouses use dimensional modeling (star schema), keep historical snapshots, and are built through ETL/ELT — distinct from a data lake, which stores raw data before it's modeled.

> *"A data warehouse is a centralized, historical repository that consolidates data from multiple source systems into a structure optimized for reporting and analysis."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Data warehouse ek central jagah hai jahan alag-alag sources (sales, HR, finance) ka data ek saath structured tareeke se store hota hai, sirf reporting/analysis ke liye — live production database pe load nahi daalta.

4. Data Modeling

53. What is a Fact Table?

ANSWER

A fact table stores measurable business events or transactions. Its columns are typically foreign keys to dimension tables plus numeric measures like quantity, sales amount, cost, or profit.

EXAMPLE

A Sales Fact table might contain DateKey, ProductKey, CustomerKey, Quantity, SalesAmount, Cost — one row per line item sold.

DEEPER DETAIL

The 'grain' of a fact table — what exactly one row represents — must be defined precisely before building it, since every downstream measure depends on that grain being consistent.

> *“A fact table stores measurable events with foreign keys to dimensions and numeric measures, and its grain defines exactly what one row means.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Fact table mein numbers/events store hote hain (jaise sales amount, quantity) plus dimensions ki foreign keys. Ye decide karna zaroori hai ki ek row exactly kya represent karti hai (grain).

54. What is a Dimension Table?

ANSWER

A dimension table holds descriptive attributes used to slice, filter and group the numbers in a fact table. Common dimensions: Customer, Product, Date, Employee, Location.

EXAMPLE

A Product dimension might contain ProductKey, ProductName, Category, Brand, Subcategory.

DEEPER DETAIL

Dimensions answer 'who, what, where, when'; facts answer 'how much/how many'. A well-designed star schema keeps these roles separated.

> *“Dimension tables hold descriptive attributes that let you slice and filter the numbers stored in fact tables.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Dimension table descriptive information rakhti hai (jaise Product ka naam, Category) jisse fact table ke numbers ko slice/filter kiya jaata hai. Facts 'kitna', dimensions 'kaun/kya/kab' batati hain.

55. What is Star Schema?

ANSWER

A star schema is a model where one central fact table connects directly to several surrounding dimension tables via single-hop relationships — no dimension joins to another dimension.

EXAMPLE

A Sales Fact table connects directly to Date, Product, Customer and Store dimensions, each linked by a single key.

DEEPER DETAIL

Contrast with snowflake schema, where dimensions are further normalized into sub-dimensions — reduces redundancy but adds extra joins, usually slower for BI tools.

> *“A star schema is a central fact table connected directly to multiple dimension tables, organized specifically for fast, simple analytical querying and reporting.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Star schema mein ek fact table beech mein hoti hai aur dimensions seedhe usse jude hote hain (ek hop) — dikhne mein star jaisa lagta hai. Isse reporting fast aur simple ho jaati hai.

56. Why do we prefer Star Schema?

ANSWER

It's simple, has clear one-hop relationships, performs well for analytical queries, and is exactly the structure Power BI's engine is optimized for. It cleanly separates facts from dimensions.

DEEPER DETAIL

Fewer joins means DAX filter propagation is simpler and more predictable — in a snowflaked model, filters travel through extra hops, which can slow things down or cause ambiguity.

> *"I prefer a star schema for reporting because it is simple, has clear relationships, and works well for analytical queries and Power BI models."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Star schema simple hota hai aur Power BI ka engine specifically isi ke liye optimize hai — kam joins matlab fast performance aur predictable filters.

57. What is a Date Table?

ANSWER

A date table is a dedicated table with one row per continuous calendar date, plus derived attributes like Year, Quarter, Month, Day, Day of Week.

EXAMPLE

Columns: Date, Year, Quarter, Month, Month Number, Day, Day of Week.

DEEPER DETAIL

It must be marked as the official 'Date Table' in Power BI and contain no gaps, or time-intelligence functions will silently produce wrong results.

> *"A date table is a dedicated, continuous calendar table with time attributes, and it's essential for accurate time-intelligence calculations."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Date table ek dedicated calendar table hoti hai (har date ke liye ek row, plus Year/Month/Quarter columns). Isme koi gap nahi hona chahiye, warna YTD/MTD jaise calculations galat ho jaayenge.

58. What is Data Normalization (with example)?

ANSWER

Normalization means organizing data across related tables so each piece of information is stored in exactly one place, reducing duplicate data and integrity problems.

EXAMPLE

Instead of repeating a customer's name/address on every order row, a normalized design stores it once in a Customers table, referenced via CustomerID.

DEEPER DETAIL

Normalization has normal forms (1NF, 2NF, 3NF) each removing a specific kind of redundancy; 3NF is commonly the target 'normalized enough' level.

> *"Data normalization means organizing data into related tables to reduce duplicate data and avoid update/insert/delete anomalies, typically by designing toward third normal form."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Normalization ka matlab hai data ko is tarah organize karna ki har information sirf ek jagah store ho, taaki data repeat na ho aur inconsistency na aaye.

5. ETL + Data Warehouse

59. What is ETL?

ANSWER

ETL stands for Extract, Transform, Load. Extract pulls data from sources; Transform cleans/validates/joins/applies business rules; Load writes prepared data into the target warehouse.

DEEPER DETAIL

ELT (Extract, Load, Transform) is a modern variant where raw data is loaded first and transformed afterward inside the warehouse, using cheap cloud compute.

> *"ETL is a process where data is extracted from sources, transformed according to business requirements, and loaded into a data warehouse."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
ETL matlab Extract (data nikaalo), Transform (usse saaf/business rules ke hisaab se badlo), Load (warehouse mein daalo). ELT ek modern variant hai jisme pehle load karke baad mein transform karte hain.

60. Full Load vs Incremental Load

ANSWER

A full load reprocesses the entire dataset every run. An incremental load processes only new or changed records since the last run.

EXAMPLE

A table with 10 million records where only 10,000 changed today: incremental load only touches those 10,000.

DEEPER DETAIL

Incremental loading needs a reliable way to identify changes — a LastModifiedDate/watermark column, CDC, or an audit table.

> *"Incremental loading processes only new or changed records to reduce processing time and resource usage, but it requires a reliable way to identify what changed."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Full load har baar poora data phir se process karta hai. Incremental load sirf naya/changed data process karta hai — zyada efficient, lekin change track karne ka tareeka chahiye hota hai.

61. What is Data Quality?

ANSWER

Data quality means data is accurate, complete, consistent, valid and reliable for its intended use. Checks include NULLs, duplicates, invalid types, key integrity, valid ranges, outliers, and row-count matching between source and target.

DEEPER DETAIL

A practical habit: source-to-target row count reconciliation after every load, so a truncated/partial load gets caught immediately.

> *"I treat data quality as a set of checks that confirm the data is accurate, complete, consistent, valid, and reliable, and I reconcile row counts between source and target after each load."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Data quality ka matlab hai data sahi, complete, aur consistent ho. Isme NULLs, duplicates, galat data types, aur source-target row count match karna check karte hain.

62. What is a Surrogate Key?

ANSWER

A surrogate key is an artificial, system-generated identifier (usually an incrementing integer) used to uniquely identify a record in the warehouse, especially in dimension tables, instead of the source system's natural key.

EXAMPLE

A Customer dimension might use Customer_SK = 1001 internally, while the source system's own ID is Customer_ID = C458.

DEEPER DETAIL

Surrogate keys make SCD Type 2 possible, since the same real-world customer can have multiple historical rows, each needing its own unique surrogate key.

> *"A surrogate key is a system-generated, warehouse-only identifier that uniquely identifies each dimension record, independent of the source system's natural key."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Surrogate key ek artificial, system-generated ID hai (jaise 1, 2, 3...) jo warehouse ke andar record ko unique identify karta hai — source system ke original ID se alag.

63. What is SCD (Slowly Changing Dimension)?

ANSWER

SCD is the data warehousing design pattern for deciding what happens when information about a customer, product, or other dimension changes over time.

DEEPER DETAIL

Several commonly discussed SCD types — 0, 1, 2, 3, 4, 6 — each answer that question differently.

> *“SCD is the data warehousing technique for deciding how to handle changes in dimension attributes over time — whether to overwrite, preserve history, or something in between.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

SCD ye decide karta hai ki jab kisi dimension (jaise customer ki city) mein change ho, to warehouse kya kare — overwrite kare, purana rakhe, ya dono. Ye ek data warehousing concept hai.

64. SCD Type 0 — Retain Original

ANSWER

In Type 0, the original value is never changed once loaded, no matter what happens in the source system afterward. The attribute is treated as fixed/immutable.

EXAMPLE

A customer's Original Signup Country is captured once and never overwritten, even if the customer relocates later.

DEEPER DETAIL

Used for attributes where the historical, original value is what matters — like first-touch attribution or original registration details.

> *“Type 0 keeps the original value permanently and never updates it, used when the initial/first value itself is what matters for reporting.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Type 0 mein original value kabhi change nahi hoti, chahe source mein kuch bhi ho jaaye — jaise customer ka pehla signup country hamesha wahi rehta hai.

65. SCD Type 1 — Overwrite

ANSWER

In Type 1, the old value is simply replaced with the new one. No history is kept.

EXAMPLE

If a customer moves from Delhi to Mumbai, the City column is updated directly — past reports re-run today show Mumbai as if it was always Mumbai.

DEEPER DETAIL

Type 1 is simplest/cheapest but historical reports can silently change meaning after the fact — fine for correcting typos, risky when historical accuracy matters.

> *“Type 1 overwrites the old value with the new one and keeps no history, which is appropriate when only the current value is needed.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Type 1 mein purani value seedhi overwrite ho jaati hai, history nahi rakhi jaati. Simple hai lekin purane reports bhi naye data se badal jaate hain.

66. SCD Type 2 — Full History (most important for interviews)

ANSWER

In Type 2, a brand-new row is inserted for the changed version, with effective-date columns and a current-record flag, preserving the complete history of every change.

EXAMPLE

If a customer moves from Delhi to Mumbai, the warehouse keeps the original Delhi row (EndDate set, IsCurrent=0) and adds a new Mumbai row (StartDate=today, IsCurrent=1).

DEEPER DETAIL

This is the SCD type interviewers focus on most — it enables accurate historical analysis. It requires a surrogate key so two versions of the same customer can coexist as separate rows.

> *“Type 2 preserves full history by inserting a new row for each change, using effective dates and a current-record flag, which is the standard approach when historical accuracy matters.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Type 2 sabse important hai — change hone par purani row rakh li jaati hai aur nayi row add ho jaati hai, StartDate/EndDate/IsCurrent flag ke saath. Isse poori history accurately track hoti hai.

67. SCD Type 3 — Limited History

ANSWER

In Type 3, the old value and current value are kept side by side as separate columns in the same row — only one prior state is retained, not a full change log.

EXAMPLE

A Customer row might have City_Current = Mumbai and City_Previous = Delhi in the same row.

DEEPER DETAIL

Cheaper/simpler than Type 2 (no row explosion), but only remembers the immediately previous value — a second change overwrites and loses the earlier value.

> *“Type 3 keeps limited history by storing the old and current values in separate columns on the same row, useful for simple before/after comparisons but not for a full change history.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Type 3 mein sirf ek purani value aur current value dono ek hi row mein alag columns mein rakhi jaati hai — poori history nahi, sirf 'pehle vs ab' compare kar sakte ho.

68. SCD Type 4 — Separate History Table

ANSWER

In Type 4, the current value stays in the main dimension table (fast, simple for everyday queries), while all historical versions move to a separate history table.

EXAMPLE

A Customer_Current table has today's value; a Customer_History table logs every past version with its own date range.

DEEPER DETAIL

Keeps the main dimension small/fast while still preserving full history — practical for very large, frequently-changing dimensions.

> *“Type 4 splits current and historical data into two separate tables, keeping the main dimension fast while still preserving full history elsewhere.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Type 4 mein current data ek table mein aur poora history ek alag table mein rakhi jaati hai — isse main table fast rehti hai aur history bhi safe rehti hai.

69. SCD Type 6 — Hybrid

ANSWER

Type 6 combines ideas from Types 1, 2 and 3: it can maintain the current value, the immediately previous value, and a full history of rows, all at once.

EXAMPLE

A dimension might have row-per-version history (Type 2), plus a Current_City column (Type 1 style), plus a Previous_City column (Type 3 style).

DEEPER DETAIL

Powerful but adds real design/ETL complexity — reserved for dimensions where different reports genuinely need different views of history.

> *“Type 6 is a hybrid approach that combines current value, previous value, and full historical rows in one design, giving maximum flexibility at the cost of extra complexity.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Type 6 Type 1, 2, aur 3 teeno ka mix hai — current value, previous value, aur poori history teeno ek saath rakhta hai. Flexible hai lekin complex bhi hai.

70. OLTP vs OLAP

ANSWER

OLTP systems are optimized for many small, fast reads and writes — day-to-day apps like order-entry. OLAP systems are optimized for complex, read-heavy analytical queries over large historical volumes, like a data warehouse.

EXAMPLE

An e-commerce checkout inserting one order at a time is OLTP. A Power BI report summarizing three years of sales trends queries an OLAP warehouse.

DEEPER DETAIL

OLTP databases are typically highly normalized (fast, safe writes); OLAP systems are typically denormalized into star schemas (fast, simple reads).

> *“OLTP systems are built for fast day-to-day transactions, while OLAP systems are built for complex analytical queries over historical data, which is why we separate them with a data warehouse.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
OLTP roz-marra ke chote-chote transactions ke liye hota hai (jaise order place karna). OLAP badi analytical queries ke liye hota hai (jaise 3 saal ka sales trend dekhna). Isliye warehouse alag rakha jaata hai.

71. What is a Staging Area in ETL?

ANSWER

A staging area is an intermediate landing zone where raw data is loaded temporarily, exactly as extracted from the source, before any transformation or validation happens.

EXAMPLE

Data is copied from the source CRM into a staging table unchanged; only after quality checks does clean data get loaded into the actual warehouse tables.

DEEPER DETAIL

Staging isolates the source from transformation logic, makes re-running a failed load easy without re-extracting, and gives a raw snapshot to debug against.

> *“A staging area holds raw extracted data temporarily before transformation, which protects the source system and makes failed loads easier to re-run and debug.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Staging area ek beech ki jagah hai jahan raw data pehle load hota hai, bina transform kiye. Isse source system safe rehta hai aur load fail hone par dobara run karna aasan ho jaata hai.

72. What is a Junk Dimension?

ANSWER

A junk dimension is a single dimension table that groups several unrelated, low-cardinality flags or indicators that don't deserve their own separate tables, keeping the fact table cleaner.

EXAMPLE

Instead of five separate flag columns (IsGift, IsTaxExempt, PaymentMethod...) on a large Sales fact table, they're combined into one small Junk Dimension.

DEEPER DETAIL

Called 'junk' because the combined attributes have no natural relationship beyond convenience — the goal is avoiding a fact table with dozens of tiny flag columns.

> *“A junk dimension bundles several unrelated low-cardinality flags into one small dimension table, keeping the fact table cleaner.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Junk dimension chote-chote unrelated flags (jaise IsGift, PaymentMethod) ko ek table mein jod deta hai, taaki fact table mein bahut saare chote columns na bikhre rahen.

73. What is a Conformed Dimension?

ANSWER

A conformed dimension is built once with a consistent structure/meaning, then shared and reused across multiple fact tables or data marts, so a dimension like Date or Customer means exactly the same everywhere.

EXAMPLE

The same Date dimension is used by both Sales and Inventory fact tables, so a report can compare sales and inventory for the same period without mismatch.

DEEPER DETAIL

Conformed dimensions make it possible to build separate data marts for different departments that can still be combined/compared reliably.

> *“A conformed dimension is shared consistently across multiple fact tables or data marts, ensuring dimensions like Date or Customer mean the same thing everywhere they're used.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Conformed dimension ek hi tarah ki dimension hai jo multiple fact tables mein reuse hoti hai — jaise ek hi Date table Sales aur Inventory dono mein use ho, taaki comparison sahi rahe.

74. What is Data Lineage?

ANSWER

Data lineage is the traceable record of where data originated, what transformations it went through, and where it ends up — the end-to-end map from source to final report.

EXAMPLE

If a Total Revenue figure looks wrong, lineage lets you trace backward: the DAX measure → model table → ETL transformation → staging table → original source column.

DEEPER DETAIL

Good lineage documentation makes root-cause debugging faster and is often required for compliance/audit purposes.

> *“Data lineage traces a piece of data from its original source through every transformation to its final destination, which makes debugging and audit requirements much easier.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Data lineage batata hai ki koi number report mein kahan se aaya — shuru se lekar end tak ka poora path (source → transformation → report). Isse debugging aasan hoti hai.

6. Python / API

75. How do you read a CSV in Python/Pandas?

ANSWER

pandas.read_csv() loads a CSV file straight into a DataFrame, the core table-like structure pandas is built around.

EXAMPLE

import pandas as pd; df = pd.read_csv("sales.csv")

DEEPER DETAIL

df.head() previews rows, df.info() shows types/nulls, df.describe() gives summary stats — the standard first three checks on a new dataset.

> *"I use pandas.read_csv() to load CSV data into a DataFrame, then inspect it with head(), info() and describe()."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
pd.read_csv() se CSV file DataFrame mein load hoti hai. Uske baad head(), info(), describe() se data ko quickly check karte hain.

76. How do you handle missing values?

ANSWER

Identify where values are missing and why, then based on business requirement remove, fill with a default, or impute using mean/median.

EXAMPLE

df.isna().sum(); df["salary"] = df["salary"].fillna(df["salary"].median())

DEEPER DETAIL

Don't blindly replace every missing value with zero — a missing value (unknown) can carry different meaning than an actual zero.

> *"I identify missing values, understand why they're missing, and then choose to remove, fill with a default, or impute based on the actual business context."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Missing values ke liye pehle pata karo kyun missing hain, phir decide karo: hatao, default value se bharo, ya median/mean se impute karo. Har missing value ko zero se replace mat karo.

77. List vs Dictionary

ANSWER

A list is an ordered collection accessed mainly by numeric position/index. A dictionary stores key-value pairs, accessed by key rather than position.

EXAMPLE

names = ["Amit", "Riya"]; employee = {"name": "Amit", "salary": 50000}

DEEPER DETAIL

Lists are ideal for ordered sequences where position matters. Dictionaries are ideal whenever each value has a meaningful label.

> *"A list is useful for ordered sequences accessed by position; a dictionary is useful when each value has a meaningful key to look it up by."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
List ordered items rakhti hai jo position (index) se access hoti hai. Dictionary key-value pairs rakhti hai, jisse aap key se value dhoondte ho.

78. What is groupby in Pandas?

ANSWER

groupby splits a DataFrame into groups based on one or more columns, then lets you run aggregate calculations like sum/count/mean on each group separately.

EXAMPLE

df.groupby("department")["salary"].sum()

DEEPER DETAIL

groupby follows split-apply-combine: split into groups, apply aggregation to each, combine results — same as SQL's GROUP BY.

> *"I use groupby to perform grouped aggregations and summarize data by business dimensions, similar to GROUP BY in SQL."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
groupby Pandas mein SQL ke GROUP BY jaisa kaam karta hai — data ko groups mein baant ke har group pe aggregation (sum, count) karta hai.

79. What is an API?

ANSWER

API stands for Application Programming Interface. It's a defined contract that lets different applications communicate, without either side needing to know the other's internal implementation.

EXAMPLE

A Python application calls a REST API to retrieve customer or weather data, typically as JSON.

DEEPER DETAIL

REST is the most common style — standard HTTP methods (GET, POST, PUT, DELETE) against resource-based URLs, returning JSON, which pandas can read with pd.json_normalize() or pd.read_json().

> *"An API is an interface that allows different software systems to communicate and exchange data or functionality."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
API ek contract hai jisse do alag software aapas mein baat kar sakte hain, bina ek-doosre ka internal code jaane. REST APIs commonly JSON data return karte hain.

80. GET vs POST

ANSWER

GET is generally used to retrieve data without changing anything. POST is generally used to send data to a server, often to create a new resource.

EXAMPLE

GET /students retrieves the list; POST /students sends new student data to create a record.

DEEPER DETAIL

GET requests are idempotent and safe to repeat/cache; POST requests are not — calling POST twice by accident could create two duplicate records.

> *"GET is mainly for retrieving data, while POST is mainly for sending data or creating a resource, and only GET is safe to repeat without side effects."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
GET data retrieve karne ke liye hota hai (kuch change nahi hota). POST naya data bhejne/create karne ke liye hota hai. GET baar baar call karna safe hai, POST nahi.

81. What is Pagination?

ANSWER

Pagination means splitting a large API response into smaller, sequential pages instead of returning every record in one response.

EXAMPLE

An API might return 100 customers per page, with parameters like page=1, page=2.

DEEPER DETAIL

When pulling paginated data, you typically loop, requesting the next page until empty or a 'has_more' field is false, then concatenate into one DataFrame.

> *"Pagination divides a large API response into smaller pages, which reduces response size and improves performance and usability."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Pagination bade data ko chote-chote pages mein baant deta hai (jaise 100 records per page), taaki ek baar mein sab kuch load na ho aur performance achhi rahe.

82. What is a Rate Limit?

ANSWER

A rate limit caps how many requests a client can make to an API within a given time window, to protect the service and keep access fair.

EXAMPLE

An API might allow only 60 requests per minute; exceeding it returns HTTP 429 'Too Many Requests'.

DEEPER DETAIL

A robust integration reads the Retry-After header and backs off with a pause before retrying, rather than hammering the API.

> *"A rate limit controls how many requests a client can make in a given time period, and I design integrations to back off and retry gracefully when they hit one."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Rate limit batata hai ki ek time period mein aap kitni requests bhej sakte ho. Limit cross karne par 429 error aata hai — isliye retry logic mein thoda wait karna chahiye.

83. How do you reverse a string in Python?

ANSWER

The simplest way is slice syntax [::-1]. It can also be done with reversed() + join, or manually with a loop.

EXAMPLE

text = "analyst"; reversed_text = text[::-1] # tsylana

DEEPER DETAIL

Interviewers sometimes ask to reverse without slicing/built-ins — the manual loop is the answer to have ready for that.

> *"The quickest way is text[::-1] using Python's slice notation with a step of -1, and I can also do it manually with a loop if asked to avoid built-ins."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
String reverse karne ka sabse aasan tareeka hai text[::-1]. Loop se manually bhi kar sakte ho agar built-in use na karna ho.

84. Pandas vs PySpark

ANSWER

Pandas runs on a single machine, suited to small/medium datasets that fit in memory. PySpark is built for distributed processing across multiple machines, designed for very large datasets.

DEEPER DETAIL

Spark has more overhead for small datasets — for a file that fits in memory, pandas will usually be faster and simpler. PySpark becomes worthwhile once data volume genuinely demands it.

> *"I would choose Pandas for data that can be handled efficiently on one machine and PySpark when distributed processing is needed for very large data."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Pandas ek machine pe kaam karta hai, chote-medium data ke liye theek hai. PySpark bade data ko multiple machines par distribute karke process karta hai.

85. Spark Transformation vs Action

ANSWER

A transformation defines an operation and builds a new DataFrame/RDD lineage lazily, without running yet. An action triggers Spark to execute the whole lineage and produce a result.

EXAMPLE

filter(), select(), groupBy() are transformations. count(), show(), collect() are actions.

DEEPER DETAIL

This 'lazy evaluation' model lets Spark optimize the entire chain of transformations together once an action finally forces execution.

> *"Transformations define what Spark should do, lazily; actions trigger execution and produce a result, which is what allows Spark to optimize the whole chain before running it."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Transformation sirf plan banata hai, turant execute nahi hota (lazy). Action asli mein us plan ko run karke result deta hai. Isse Spark poore chain ko optimize kar sakta hai.

86. How do you merge/join two DataFrames in Pandas?

ANSWER

pandas.merge() combines two DataFrames based on shared key columns, working like a SQL JOIN, with an explicit how parameter for join type.

EXAMPLE

merged = pd.merge(orders, customers, on="customer_id", how="left")

DEEPER DETAIL

The how parameter mirrors SQL: 'inner' keeps matches, 'left' keeps all of left, 'right' keeps all of right, 'outer' keeps everything, filling NaN for unmatched.

> *"pd.merge() joins DataFrames on a shared key, and the how parameter (inner, left, right, outer) works exactly like the equivalent SQL JOIN type."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
pd.merge() Pandas mein SQL JOIN jaisa kaam karta hai — do DataFrames ko common key pe jodta hai, aur how parameter se join type (inner/left/right/outer) choose karte ho.

87. apply() vs map() vs applymap() in Pandas

ANSWER

map() applies a function element-wise on a single Series. apply() is more flexible and can run row-wise/column-wise across a DataFrame. applymap() applies element-wise across every cell of a DataFrame.

EXAMPLE

```
df["dept_upper"] = df["department"].map(str.upper); df["bonus"] = df.apply(lambda row: row["salary"]*0.1 if row["department"]=="Sales" else 0, axis=1)
```

DEEPER DETAIL

apply() with axis=1 processes row by row, slower than a vectorized operation on large DataFrames — prefer native vectorized expressions when available.

> *“I use map() for simple single-column lookups or transforms, and apply() when I need row-wise or more complex logic, while preferring native vectorized operations for performance.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

map() ek column pe simple transform karta hai. apply() zyada flexible hai, row-wise ya complex logic ke liye. applymap() poore DataFrame ke har cell pe apply hota hai.

88. How do you remove duplicate rows in Pandas?

ANSWER

drop_duplicates() removes exact duplicate rows by default, or duplicates based on a specific column subset if passed.

EXAMPLE

```
df.drop_duplicates(subset=["customer_id"], keep="first", inplace=True)
```

DEEPER DETAIL

The keep parameter controls which duplicate survives: 'first', 'last', or False to drop all copies. df.duplicated().sum() checks how many exist first.

> *“drop_duplicates() removes duplicate rows, and I use the subset and keep parameters to control exactly which column combination defines a duplicate.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

drop_duplicates() duplicate rows hata deta hai. subset parameter se decide karte ho kaunse column(s) ke basis pe duplicate maana jaaye, aur keep se kaunsa copy rakhna hai.

89. What is a virtual environment in Python and why use one?

ANSWER

A virtual environment is an isolated Python installation with its own set of installed packages, separate from system-wide Python and other projects, so different projects can use different package versions without conflict.

EXAMPLE

```
python -m venv myenv; source myenv/bin/activate; pip install pandas
```

DEEPER DETAIL

Without virtual environments, one project's required pandas version can silently break another project needing a different version.

> *“A virtual environment isolates a project's package dependencies from the system Python and other projects, which avoids version conflicts.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Virtual environment ek alag, isolated Python setup hota hai har project ke liye — taaki alag-alag projects mein alag package versions use ho sakein, bina conflict ke.

90. Mutable vs Immutable data types in Python

ANSWER

A mutable object can be changed in place after creation. An immutable object cannot be changed once created — any 'modification' creates a brand-new object.

EXAMPLE

Lists, dicts, sets are mutable. Strings, tuples, integers are immutable — text.upper() returns a new string, doesn't change the original.

DEEPER DETAIL

This matters for function arguments: passing a mutable object into a function and modifying it changes the original outside the function too — a subtle bug source.

> *“Mutable objects like lists and dictionaries can be changed in place; immutable objects like strings and tuples can't — any change creates a new object instead.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Mutable objects (list, dict) ko change kar sakte ho without naya object banaye. Immutable objects (string, tuple) change nahi ho sakte — koi bhi 'change' asal mein naya object banata hai.

7. Microsoft Fabric

91. What is Microsoft Fabric?

ANSWER

Microsoft Fabric is an end-to-end analytics platform that brings data engineering, data integration, data warehousing, data science, real-time analytics, and Power BI together under one unified SaaS platform.

DEEPER DETAIL

Power BI on its own mainly focuses on reporting/visualization. Fabric covers the entire lifecycle upstream, and Power BI is just one of Fabric's built-in workloads.

> *"Microsoft Fabric is a unified analytics platform that brings multiple data and analytics workloads, including Power BI, together in one place."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Microsoft Fabric ek unified platform hai jisme data engineering, warehousing, data science, aur Power BI sab ek jagah aa jaate hain — Power BI ab Fabric ka ek hissa hai.

92. What is OneLake?

ANSWER

OneLake is Fabric's single, centralized data lake that all Fabric workloads share, giving the organization one common storage foundation.

DEEPER DETAIL

Every Fabric workspace automatically gets a place in OneLake; data stored there (Delta/Parquet format) can be accessed by Lakehouses, Warehouses, notebooks and Power BI without duplication — often called 'OneDrive for data.'

> *"OneLake is Fabric's centralized data lake that provides a single data foundation shared across all Fabric workloads."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
OneLake Fabric ka ek central data lake hai jisse saare workloads (Lakehouse, Warehouse, Power BI) same data share karte hain — isse 'OneDrive for data' bhi kehte hain.

93. Lakehouse vs Data Warehouse in Fabric

ANSWER

A Lakehouse blends data lake flexibility with warehouse structure, suited to flexible storage plus engineering/data science workloads. A Fabric Warehouse is built specifically for structured relational data and full SQL-based analytics.

DEEPER DETAIL

Rule of thumb: reach for a Lakehouse for raw/semi-structured files needing Spark/notebook flexibility; reach for a Warehouse when data is already structured and the primary interface is SQL.

> *"I'd consider a Lakehouse when flexible data storage and engineering are important, while a Fabric Warehouse is a strong choice for structured relational and SQL-based analytics."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Lakehouse flexible hota hai (raw/semi-structured data, Spark ke liye achha). Warehouse structured relational data aur SQL-based analytics ke liye achha hai.

94. What is a Fabric Pipeline?

ANSWER

A Fabric pipeline automates and orchestrates data movement and processing tasks — extracting, transforming, loading data, and running the whole sequence on a schedule.

DEEPER DETAIL

Conceptually similar to Azure Data Factory pipelines (shares much of the same engine) — useful to mention if you have ADF experience.

> *"A Fabric pipeline automates and orchestrates data workflows so extraction, transformation, and loading can run in a controlled, scheduled sequence."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Fabric Pipeline data movement aur processing ko automate/schedule karta hai — ETL steps ko ek order mein chalata hai, bina manually run kiye.

95. What are Fabric's core workload types?

ANSWER

Data Factory (ingestion), Synapse Data Engineering (Lakehouses/Spark notebooks), Synapse Data Warehousing (SQL warehouses), Synapse Data Science (ML), Real-Time Intelligence (streaming), and Power BI (reporting).

DEEPER DETAIL

All these workloads read/write through the same underlying OneLake, which is exactly what makes Fabric feel unified rather than stitched-together products.

> *“Fabric organizes its capabilities into workload experiences — Data Factory, Data Engineering, Data Warehousing, Data Science, Real-Time Intelligence and Power BI — all built on the same OneLake foundation.”*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Fabric ke core workloads hain: Data Factory, Data Engineering, Data Warehousing, Data Science, Real-Time Intelligence, aur Power BI — sab ek hi OneLake pe based hain.

8. Communication / Project Questions

96. Explain your project in 2 minutes

ANSWER

A tight, structured walkthrough of one real project: where data came from, how you cleaned/modeled it, what calculations you built, what the report looked like, and what business outcome it enabled.

EXAMPLE

"I worked on a data analytics project... SQL for extraction, Power Query for cleaning, star schema with fact/dimension tables, DAX measures like Total Sales/YTD/Growth %, an interactive Power BI report, published with refresh configured."

DEEPER DETAIL

Substitute your own real project details — concrete table names, actual measures, real numbers always land better than a generic script.

> *"Tip: while speaking, tell your actual project's story rather than reciting this answer word-for-word."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Apna project batate waqt structure follow karo: data kahan se aaya, kaise clean/model kiya, kaunse DAX measures banaye, report kaisa dikhta tha, aur business ko kya fayda hua. Apni asli project details use karo, generic script mat bolo.

97. What if the report shows wrong numbers?

ANSWER

Compare the Power BI result against the source or an equivalent SQL query, then work through relationships, active filters/slicers, column data types, duplicate rows, and finally the DAX calculation — building test measures if needed.

> *"I troubleshoot from source to report: Source → Relationships → Filters → Data Types → DAX → Visual, and compare results with SQL where possible."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Galat numbers dikhne par seedha DAX mat badlo. Pehle source se compare karo, phir relationships, filters, data types check karo, aur last mein DAX formula dekho.

98. What challenges did you face?

ANSWER

A strong, honest example is data quality issues — NULLs, duplicates, inconsistent values — handled during transformation in SQL and Power Query. Another is a visual showing an unexpected number, resolved by checking relationships/filter context/DAX.

> *"One challenge was inconsistent data quality, which I handled during transformation using SQL and Power Query. When visuals showed unexpected numbers, I checked relationships, filters, and DAX calculations."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Ek achha jawaab hai: data quality issues (NULLs, duplicates) jo SQL/Power Query mein clean kiye. Ya kabhi visual mein galat number aaya to relationships/filters/DAX check karke fix kiya.

99. How do you explain a technical concept to a non-technical stakeholder?

ANSWER

Lead with the business impact, not the mechanism — what the number means and why it matters — using a simple analogy before going into technical detail, only if asked.

EXAMPLE

Instead of 'I applied SCD Type 2 with surrogate keys,' say 'the report keeps a full history, so if a customer changes region, we can still see accurately where their past sales happened.'

> *"I translate the technical mechanism into its business impact first, using a simple analogy, and only go deeper into the technical detail if the stakeholder asks for it."*

HINDI MEIN SAMJHAAIYE (HINGLISH)
Non-technical logo ko samjhaate waqt pehle business impact batao (kya fayda hoga), technical mechanism nahi. Simple analogy use karo, aur detail sirf tab do jab wo poochein.

100. How do you prioritize multiple report requests from different stakeholders?

ANSWER

Clarify the business impact and deadline of each request first, then align priority with whoever owns the reporting roadmap rather than deciding unilaterally, and communicate realistic timelines back to each requester.

DEEPER DETAIL

This question really tests stakeholder-management maturity — interviewers want to hear you communicate trade-offs, not silently work through a queue.

> *"I prioritize based on business impact and deadlines, confirm priority with the report owner when requests conflict, and keep every requester updated on realistic timelines."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

Multiple requests aane par pehle har ek ka business impact aur deadline samjho, phir report owner se priority confirm karo, aur sabko realistic timeline batao — khud akele decide mat karo.

9. Bonus: Data Cleaning — Python, SQL & Power BI (Side-by-Side)

A. Python (pandas) — Step by Step

DETAILS

Same messy table cleaned three different ways — the logic is identical, only the tool changes.

Sample dirty data — Table: `customers_raw`

customer_id	name	email	signup_date	city	phone
101	Amit Sharma	amit@x.com	2024-01-15	Delhi	98765xxxxx
101	Amit Sharma	amit@x.com	2024-01-15	Delhi	98765xxxxx
102	riya patel	RIYA@Y.COM	15-02-2024	delhi	NULL
103	Neha	neha@z.com	2024/03/10	Mumbai	91234xxxxx
104	NULL	null	2024-04-05	NULL	99999xxxxx

Issues: duplicate row, inconsistent casing, mixed date formats, inconsistent city spelling, NULLs.

```python
import pandas as pd

df = pd.read_csv("customers_raw.csv")

# 1. Inspect
df.info()
df.isna().sum()

# 2. Remove exact duplicates
df = df.drop_duplicates()

# 3. Standardize text case
df["name"] = df["name"].str.strip().str.title()
df["email"] = df["email"].str.strip().str.lower()
df["city"] = df["city"].str.strip().str.title()

# 4. Fix inconsistent date formats
df["signup_date"] = pd.to_datetime(df["signup_date"], errors="coerce", dayfirst=True)

# 5. Handle missing values
df["name"] = df["name"].fillna("Unknown")
df["phone"] = df["phone"].fillna("Not Provided")
df = df.dropna(subset=["email"])  # drop rows with no email

# 6. Validate data types
df["customer_id"] = df["customer_id"].astype(int)

# 7. Remove duplicates by business key
df = df.drop_duplicates(subset=["customer_id"], keep="first")
```

HINDI MEIN SAMJHAAIYE (HINGLISH)

Ye teeno tools (Python, SQL, Power BI) same kaam karte hain — duplicate hatana, text ko standard banana, date format theek karna, aur missing values handle karna. Bas tarika (syntax/UI) alag hai. Pandas mein steps: pehle inspect karo (info/isna), phir drop_duplicates() se duplicates hatao, .str.title()/.lower() se text standardize karo, pd.to_datetime() se date format fix karo, fillna()/dropna() se missing values handle karo, aur astype() se data type sahi karo.

B. SQL — Step by Step

DETAILS

```
-- 1. Find duplicates
SELECT customer_id, COUNT(*)
FROM customers_raw
GROUP BY customer_id
HAVING COUNT(*) > 1;

-- 2. Remove exact duplicate rows (keep lowest ctid per group)
DELETE FROM customers_raw a
USING customers_raw b
WHERE a.ctid < b.ctid
  AND a.customer_id = b.customer_id;

-- 3. Standardize text
UPDATE customers_raw
SET name = INITCAP(TRIM(name)),
    email = LOWER(TRIM(email)),
    city  = INITCAP(TRIM(city));

-- 4. Handle NULLs with defaults
UPDATE customers_raw
SET phone = COALESCE(phone, 'Not Provided'),
    name  = COALESCE(name, 'Unknown');

-- 5. Fix/standardize date column (Postgres example)
UPDATE customers_raw
SET signup_date = TO_DATE(signup_date::text, 'YYYY-MM-DD')
WHERE signup_date IS NOT NULL;

-- 6. Remove rows missing a critical field
DELETE FROM customers_raw WHERE email IS NULL;
```

HINDI MEIN SAMJHAAIYE (HINGLISH)
SQL mein cleaning steps: pehle GROUP BY + HAVING se duplicates dhundo, DELETE se hatao, UPDATE + TRIM/INITCAP/LOWER se text standardize karo, COALESCE se NULLs ko default value do, aur zaroori column (jaise email) missing hone par woh row DELETE kar do.

C. Power BI (Power Query / M) — Step by Step

DETAILS

Steps in Power Query editor (each becomes an "Applied Step"):

1. **Remove Duplicates** — right-click `customer_id` column → Remove Duplicates
2. **Trim & Clean text** — select `name`, `email`, `city` → Transform → Format → Trim, Clean
3. **Change Case** — `name`, `city` → Format → Capitalize Each Word; `email` → Format → lowercase
4. **Change data type** — `signup_date` column → set type to Date (set correct Locale via "Using Locale" for mixed formats)
5. **Replace Values** — replace text `null` strings with actual blank/null
6. **Replace Errors / Fill** — for `name`, `phone` — right-click → Replace Values → null → "Unknown" / "Not Provided"
7. **Filter Rows** — filter out rows where `email` is null (critical field)
8. **Remove Rows > Remove Duplicates** based on `customer_id` only, if exact match isn't guaranteed

Equivalent M code:

```
= Table.RemoveDuplicates(Source, {"customer_id"})
= Table.TransformColumns(Source, {
    {"name", Text.Proper, type text},
    {"email", Text.Lower, type text},
    {"city", Text.Proper, type text}
  })
= Table.TransformColumnTypes(Source, {{"signup_date", type date}})
= Table.ReplaceValue(Source, null, "Unknown", Replacer.ReplaceValue, {"name"})
```

HINDI MEIN SAMJHAAIYE (HINGLISH)
Power BI mein ye sab kaam Power Query editor mein visually hota hai — har action ek “Applied Step” ban jaata hai. Duplicates hatao, Trim/Clean se text saaf karo, Capitalize/Lowercase se case fix karo, date ka data type set karo, aur null values ko Replace Values se handle karo. Peeche M language ka code apne aap generate hota hai.

D. PySpark — Step by Step

Same messy table as before, cleaned using PySpark DataFrame operations. Logic mirrors the pandas and SQL versions — only the syntax changes, and PySpark scales the same steps across a distributed cluster.

SAMPLE DIRTY DATA — TABLE: customers_raw

customer_id	name	email	signup_date	city	phone
101	Amit Sharma	amit@x.com	2024-01-15	Delhi	98765xxxxx
101	Amit Sharma	amit@x.com	2024-01-15	Delhi	98765xxxxx
102	riya patel	RIYA@Y.COM	15-02-2024	delhi	NULL
103	Neha	neha@z.com	2024/03/10	Mumbai	91234xxxxx
104	NULL	null	2024-04-05	NULL	99999xxxxx

Issues: duplicate row, inconsistent casing, mixed date formats, inconsistent city spelling, NULLs.

```python
from pyspark.sql import SparkSession
from pyspark.sql import functions as F

spark = SparkSession.builder.appName("DataCleaning").getOrCreate()

df = spark.read.csv("customers_raw.csv", header=True, inferSchema=True)

# 1. Inspect
df.printSchema()
df.show(truncate=False)
df.select([F.count(F.when(F.col(c).isNull(), c)).alias(c) for c in df.columns]).show()

# 2. Remove exact duplicates
df = df.dropDuplicates()

# 3. Standardize text case
df = df.withColumn("name", F.initcap(F.trim(F.col("name")))) \
       .withColumn("email", F.lower(F.trim(F.col("email")))) \
       .withColumn("city", F.initcap(F.trim(F.col("city"))))

# 4. Fix inconsistent date formats (coalesce across possible input patterns)
df = df.withColumn(
    "signup_date",
    F.coalesce(
        F.to_date("signup_date", "yyyy-MM-dd"),
        F.to_date("signup_date", "dd-MM-yyyy"),
        F.to_date("signup_date", "yyyy/MM/dd")
    )
)

# 5. Handle missing values
df = df.withColumn("name", F.when(F.col("name").isNull(), "Unknown").otherwise(F.col("name"))) \
       .withColumn("phone", F.when(F.col("phone").isNull(), "Not Provided").otherwise(F.col("phone")))
df = df.filter(F.col("email").isNotNull())  # drop rows with no email

# 6. Validate data types
df = df.withColumn("customer_id", F.col("customer_id").cast("int"))

# 7. Remove duplicates by business key (keep first occurrence per customer_id)
from pyspark.sql.window import Window
w = Window.partitionBy("customer_id").orderBy(F.col("signup_date").asc_nulls_last())
df = df.withColumn("rn", F.row_number().over(w)).filter(F.col("rn") == 1).drop("rn")
```

> *"PySpark cleaning follows the exact same checklist as pandas — inspect, deduplicate, standardize text, fix dates, handle NULLs, validate types — but every operation runs as a distributed transformation instead of an in-memory one."*

HINDI MEIN SAMJHAAIYE (HINGLISH)

PySpark mein bhi wahi cleaning steps hote hain jo pandas aur SQL mein the — bas syntax distributed DataFrame ke hisaab se hota hai. printSchema() aur isNull() count se pehle data inspect karo, dropDuplicates() se exact duplicates hatao, initcap()/lower()/trim() se text standardize karo, to_date() se mixed date formats fix karo (coalesce se multiple formats try karo), missing values ke liye when().otherwise() ya filter() use karo, cast() se data type sahi karo, aur agar business key (customer_id) pe duplicate chahiye to Window + row_number() use karo — jaise SQL mein ROW_NUMBER() OVER() hota hai.

DEEPER DETAIL

Unlike pandas, PySpark transformations (withColumn, filter, dropDuplicates) are lazy — nothing actually runs until an action like show(), count(), or write() triggers execution across the cluster. This is why the same cleaning script that takes milliseconds to "define" in code may take longer to "run" — Spark is planning and optimizing the whole chain before executing it.

HINDI MEIN SAMJHAAIYE (HINGLISH)

Pandas turant kaam karta hai (eager), lekin PySpark ke transformations (withColumn, filter, dropDuplicates) lazy hote hain — jab tak show(), count(), ya write() jaisa action na aaye, kuch execute nahi hota. Isliye code likhte waqt "fast" lagta hai, lekin asli processing action call hone par hoti hai — Spark poore chain ko pehle optimize karta hai.

Priority Guide For Interview Preparation

Priority 1 — Must Know

Power BI basics; Power Query; Report vs Dashboard; Import vs DirectQuery; Measure vs Calculated Column; SUM vs SUMX; CALCULATE; Row vs Filter Context; YTD; SQL JOIN; WHERE vs HAVING; GROUP BY; Primary/Foreign Key; ROW_NUMBER; RANK vs DENSE_RANK; Self Join; Fact vs Dimension; Star Schema; Normalization; ETL; Full vs Incremental Load; SCD Type 1 vs Type 2; Indexing; Data Warehousing.

Hinglish: Ye topics sabse zyada poochhe jaate hain — inhe sabse pehle aur sabse achhi tarah taiyaar karo.

Priority 2

ALL / ALLEXCEPT / ALLSELECTED; Context Transition; Running Total; MoM Growth; Finding duplicates; Subquery; Data quality checks; Surrogate key; SCD Types 0, 3, 4, 6; RLS; DAX Variables; Time Intelligence functions; Window functions (LAG/LEAD); CTEs; Views vs Stored Procedures vs Triggers; OLTP vs OLAP; Staging area; Conformed/Junk dimensions; Power BI Gateway; Bookmarks vs Drill-through; Calculation groups; Report performance optimization.

Hinglish: Ye thode advanced topics hain — senior ya thoda experienced role ke liye poochhe jaate hain.

Priority 3 — JD / Fabric / Python

Microsoft Fabric; OneLake; Lakehouse vs Warehouse; Fabric Pipeline; Pandas; reverse a string in Python; API basics; GET vs POST; Pagination; Rate limits; PySpark; Spark transformation vs action; pd.merge(); apply/map/applymap; drop_duplicates; Virtual environments; Mutable vs Immutable types.

Hinglish: Agar job description mein Fabric ya Python mention hai, to ye zaroor tayyar karo.

Final Interview Tip

Keep this single flow in your head as the spine of almost every answer: **Source → SQL → ETL → Model (Star Schema) → DAX → Power BI → Publish.** Nearly every question in this guide is really just asking you to zoom into one link of that chain.

Hinglish: Interview mein ghabrao mat — jo bhi sawaal aaye, use is ek chain mein fit karke socho: Source se lekar Publish tak. Concepts ko samjho, examples practice karo, aur one-liners ko apni bhasha mein bolna seekho — ratna mat.